

Add vectors: magnitude & direction to component

Q. The magnitude and direction angle of $\vec{v}$ and $\vec{w}$ are given below in the form (r, θ) where r is the magnitude and θ is the direction angle.

$$\vec{v} = (3, 340^\circ)$$

$$\vec{w} = (6, 80^\circ)$$

Find the component form of $\vec{v} + \vec{w}$.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

~~It's okay~~

Sol

$$\begin{aligned}\vec{v} + \vec{w} &= (v_x + w_x, v_y + w_y) \\ &= (3 \cos(340^\circ) + 6 \cos(80^\circ), 3 \sin(340^\circ) + 6 \sin(80^\circ)) \\ &\approx (2.82 + 1.04, -1.03 + 5.91) \\ &\approx (3.86, 4.88)\end{aligned}$$

Q. The magnitude of $\vec{v}$ is 7, and its direction angle is 310° . The magnitude of $\vec{w}$ is 3, and its direction angle is 70° . Find the component form of $\vec{v} + \vec{w}$.

$$\begin{aligned}\vec{v} &= (7, 310^\circ) \\ \vec{w} &= (3, 70^\circ)\end{aligned}$$

Sol

$$\vec{v} + \vec{w} = (v_x + w_x, v_y + w_y)$$

for $\vec{v} = (7, 310^\circ)$,

$$v_x = 7 \cos 310^\circ \approx 4.50, \quad v_y = 7 \sin 310^\circ \approx -5.36$$

for $\vec{w} = (3, 70^\circ)$

$$w_x = 3 \cos 70^\circ \approx 1.03, \quad w_y = 3 \sin 70^\circ \approx 2.82$$

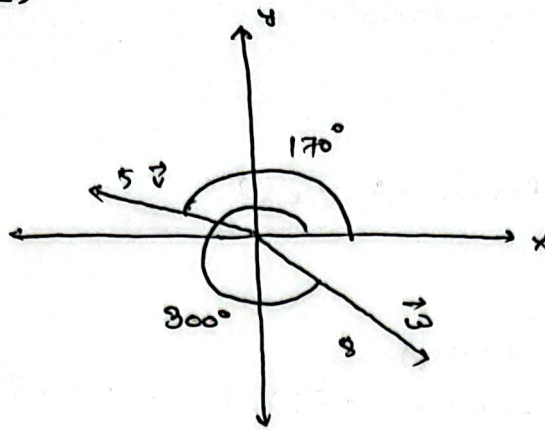
Adding the corresponding components gives

$$\begin{aligned}\vec{v} + \vec{w} &\approx (4.50 + 1.03, -5.36 + 2.82) \\ &= (5.53, -2.54)\end{aligned}$$

Q. $\vec{v}$ and $\vec{w}$ are shown below

Find the component form of $\vec{v} + \vec{w}$.

$\vec{v} + \vec{w} = (\quad , \quad)$



$$\vec{v} = (5, 170^\circ)$$

$$\vec{w} = (8, 300^\circ)$$

Sol

for $\vec{v} = (5, 170^\circ)$,

$$v_x = 5 \cos 170^\circ \approx -4.92, \quad v_y = 5 \sin 170^\circ \approx 0.87$$

for $\vec{w} = (8, 300^\circ)$,

$$w_x = 8 \cos 300^\circ \approx 4, \quad w_y = 8 \sin 300^\circ \approx -6.93$$

$$\begin{aligned} \vec{v} + \vec{w} &= (-4.92 + 4, 0.87 + (-6.93)) \\ &= (-0.92, -6.06) \end{aligned}$$

Q. for, $\vec{v} = (3, 200^\circ)$

$\vec{w} = (5, 70^\circ)$

for $\vec{v} = (3, 200^\circ)$,

$$v_x = 3 \cos 200^\circ \approx -6.58, \quad v_y = 3 \sin 200^\circ \approx -2.39$$

for $\vec{w} = (5, 70^\circ)$

$$w_x = 5 \cos 70^\circ \approx 1.71, \quad w_y = 5 \sin 70^\circ \approx 4.70$$

$$\begin{aligned} \vec{v} + \vec{w} &= (-6.58 + 1.71, -2.39 + 4.70) \\ &= -4.87, 2.30 \end{aligned}$$